Improving Evolutionary Van Gogh

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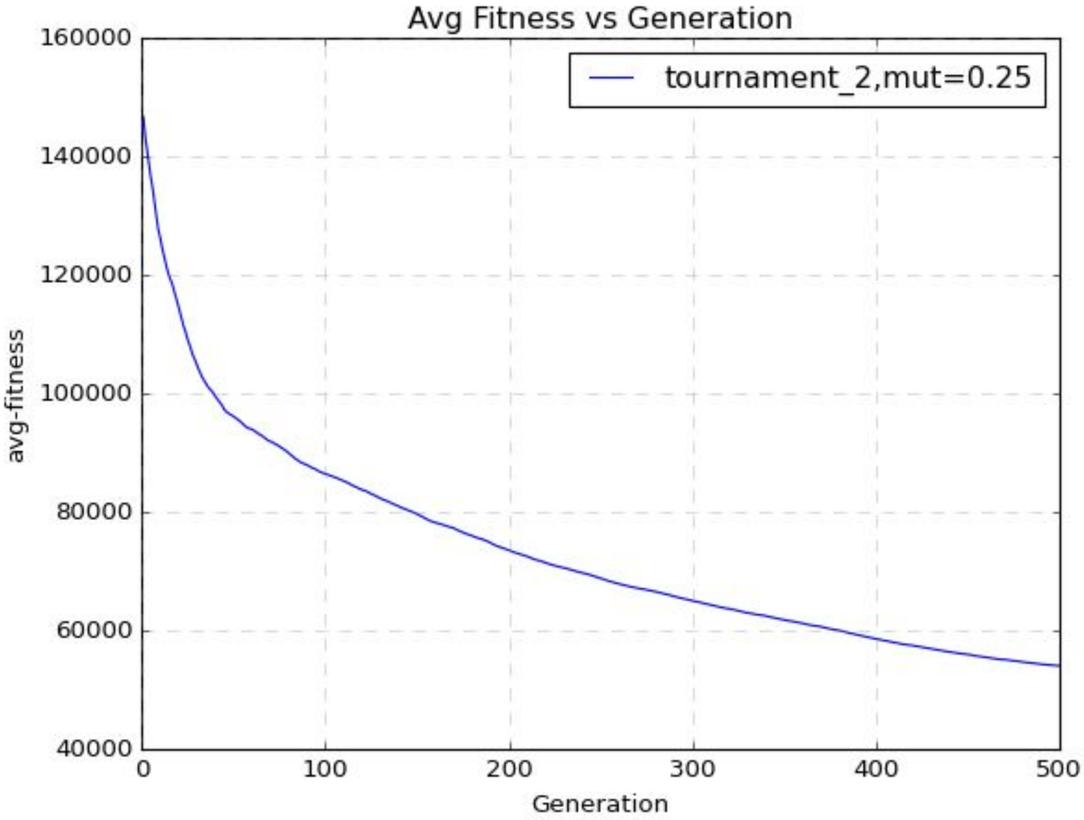
Group 16

Selection Methods to improve avg-fitness

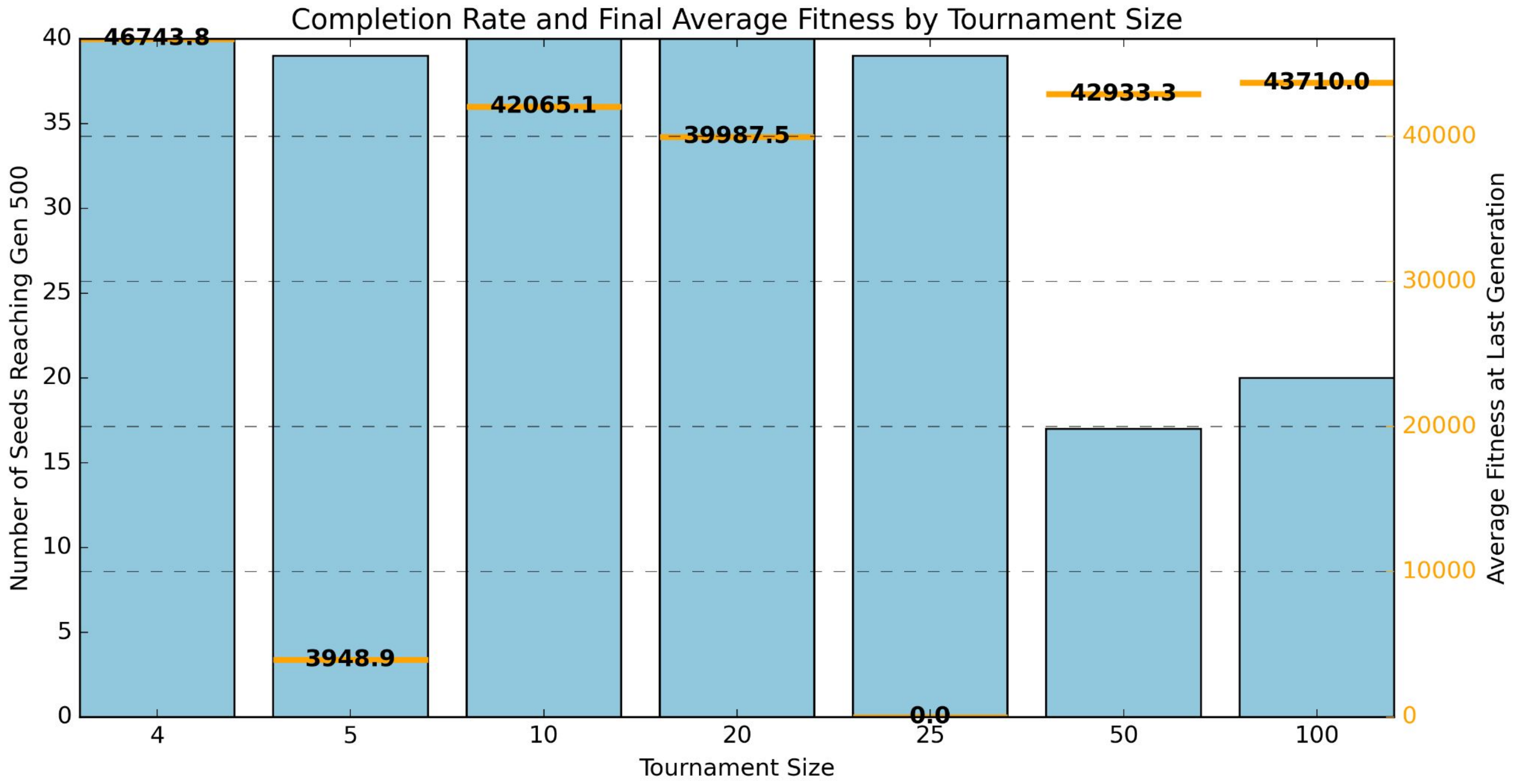


Selection Methods to improve avg-fitness

Research Question

Which selection methods and hyperparameter configuration maximizes the reduction in average population fitness over 500 generations compared to the initial method and parameters: tournament size 4 and mutation strength 0.25?

Selection Methods to improve avg-fitness : Tournament



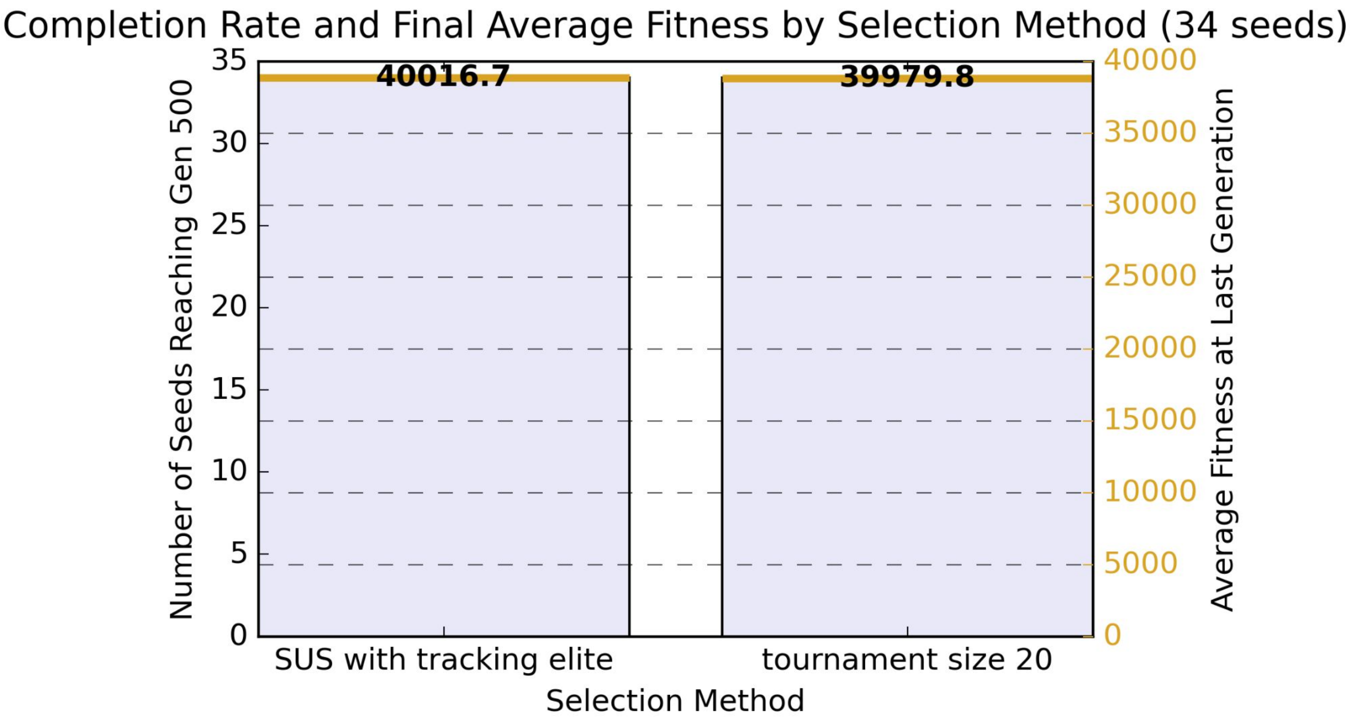
Selection Methods to improve avg-fitness: Strategy

tournament size 20 -> truncation

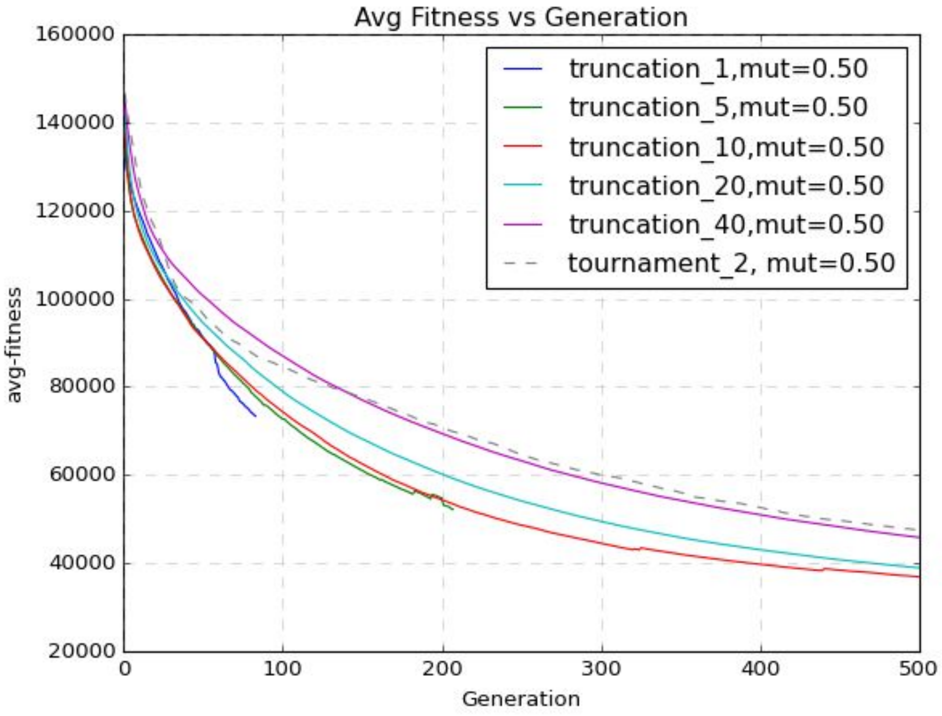
challenge: proportionate selection -> Stochastic Uniform Sampling (SUS)

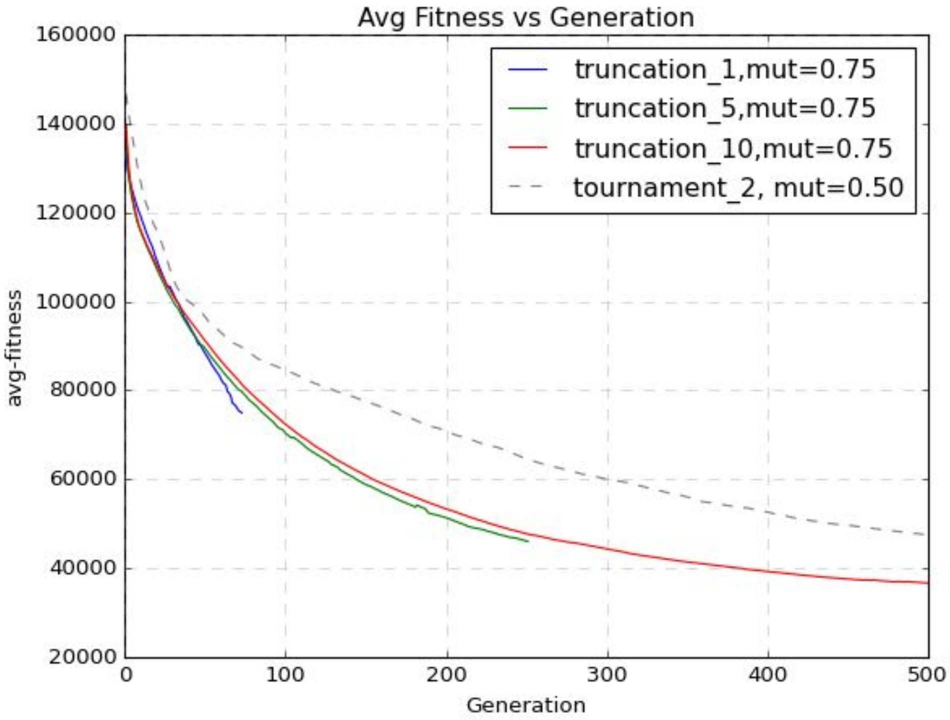
Elitism!

Selection Methods to improve avg-fitness : Tournament and SUS

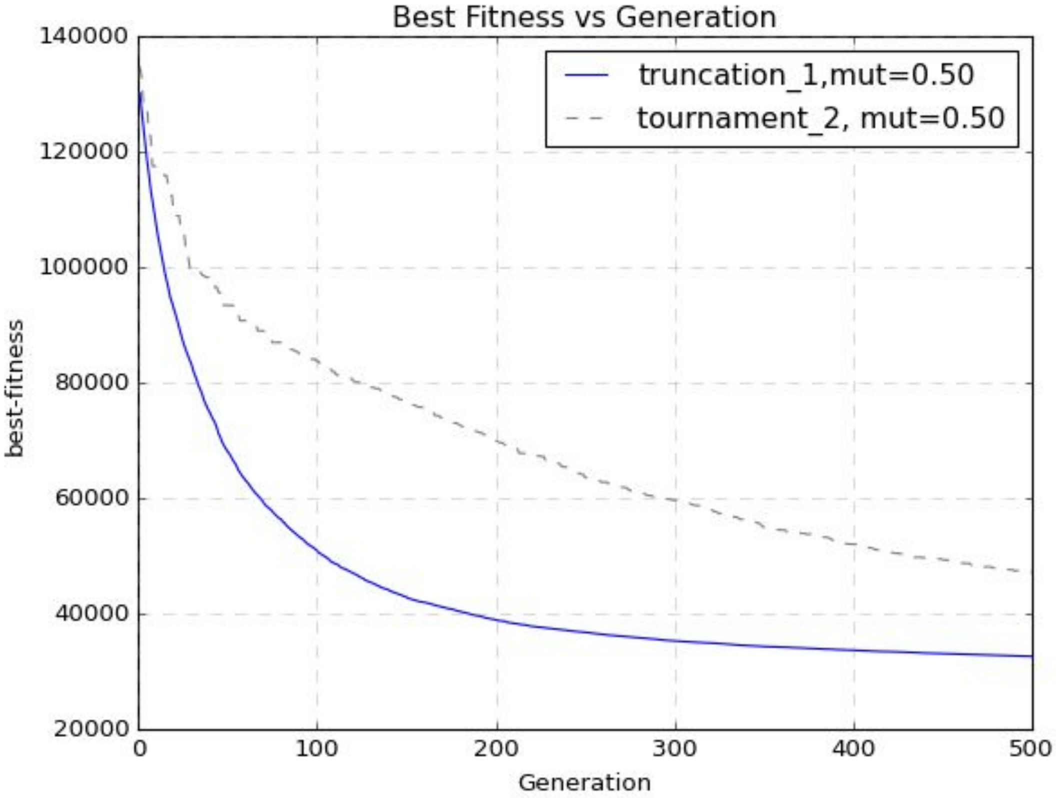


Selection Methods to improve avg-fitness : Truncation





Selection Methods to improve avg-fitness : SSGA



Preserving Voronoi-points during Variation

A single Voronoi diagram is represented as:

xyrgbxyrgbxyrgbxyrgbxyrgbxyrgb

What would a good schema look like?

How about: ⋆⋆⋆⋆⋆xy⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆?

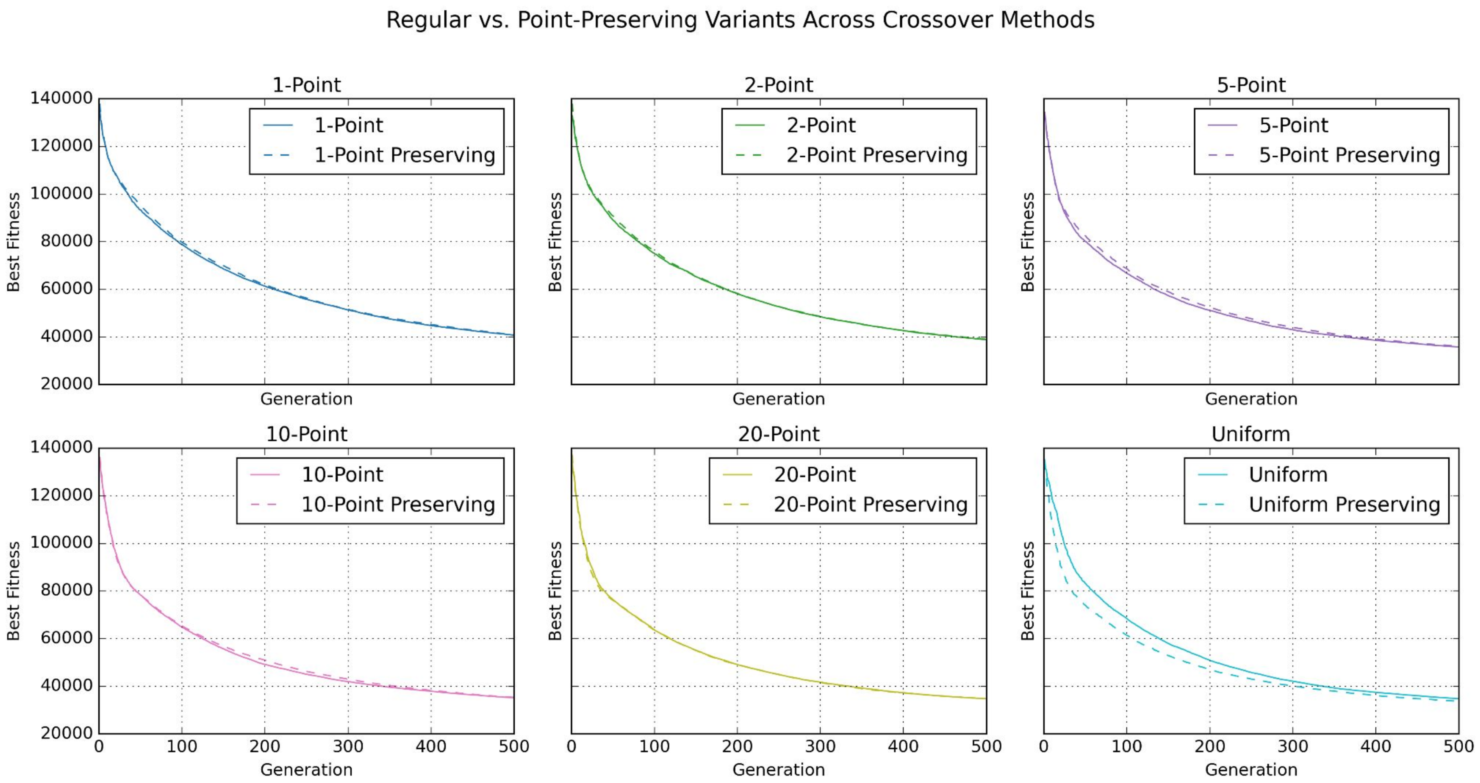
And what about: ⋆⋆⋆⋆⋆xyrgb⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆⋆?

Or: ⋆⋆⋆⋆⋆xyrgbxyrgb⋆⋆⋆⋆⋆xyrgb⋆⋆⋆⋆⋆?

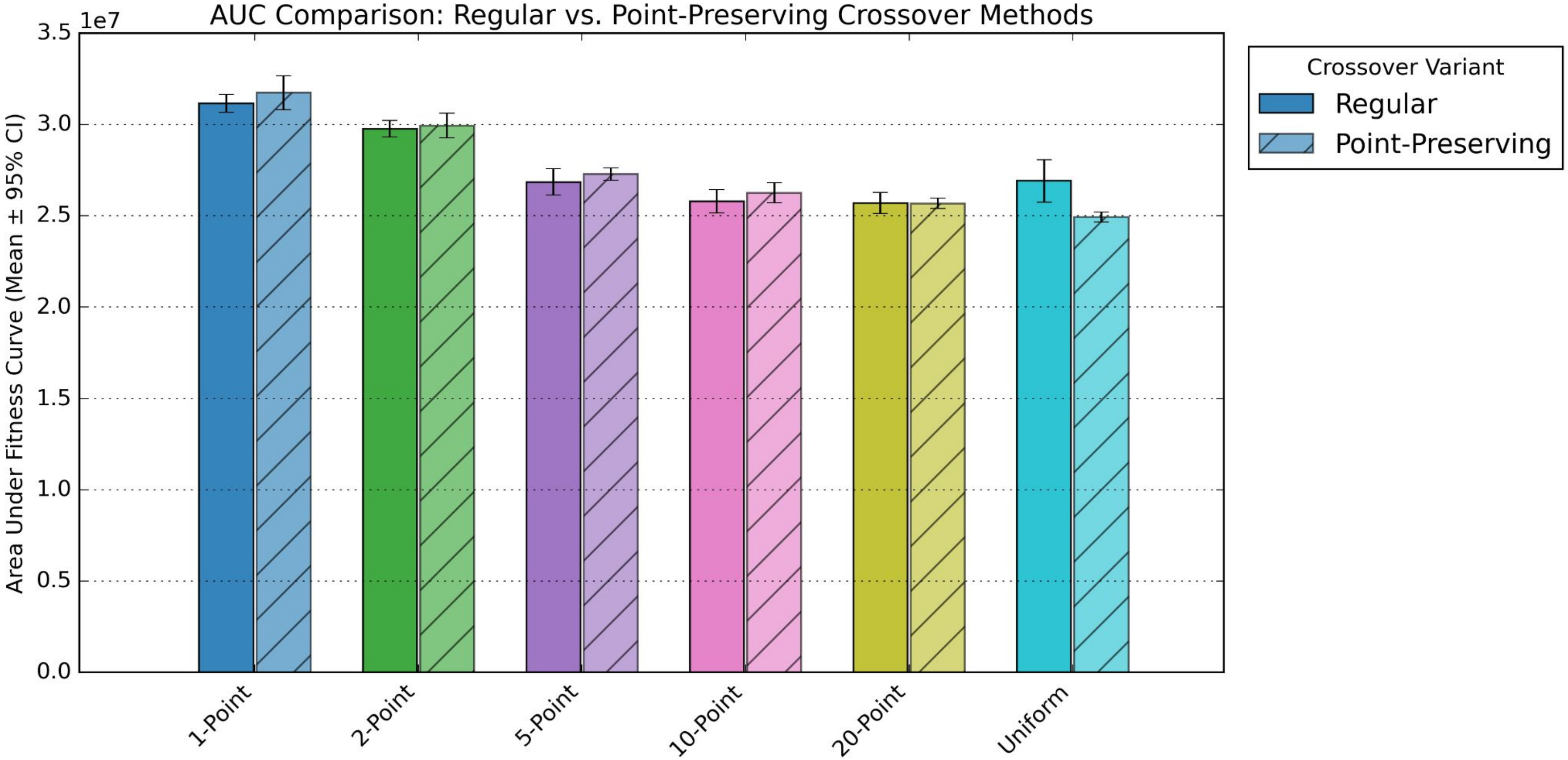
Preserving Voronoi-points during Variation

How does preserving the (x, y, r, g, b)-structure of individual Voronoi points during crossover affect the algorithm's convergence rate?

Preserving Voronoi-points during Variation



Preserving Voronoi-points during Variation: Results



Grey-box Initialization - Why?

Problem: Slow Convergence in Evolutionary Models

- The model takes a long time to converge
- It tries many random colors and positions.
- Final fitness after generations varies depending on the initial population.
- Results are inconsistent and unpredictable across runs.
- We suspect that better initialization could improve convergence and stability.

Grey-box Initialization - Using Domain Knowledge

Research question: To what extent does initializing the population using points extracted from reference image improve convergence speed and solution quality compared to random initialization?

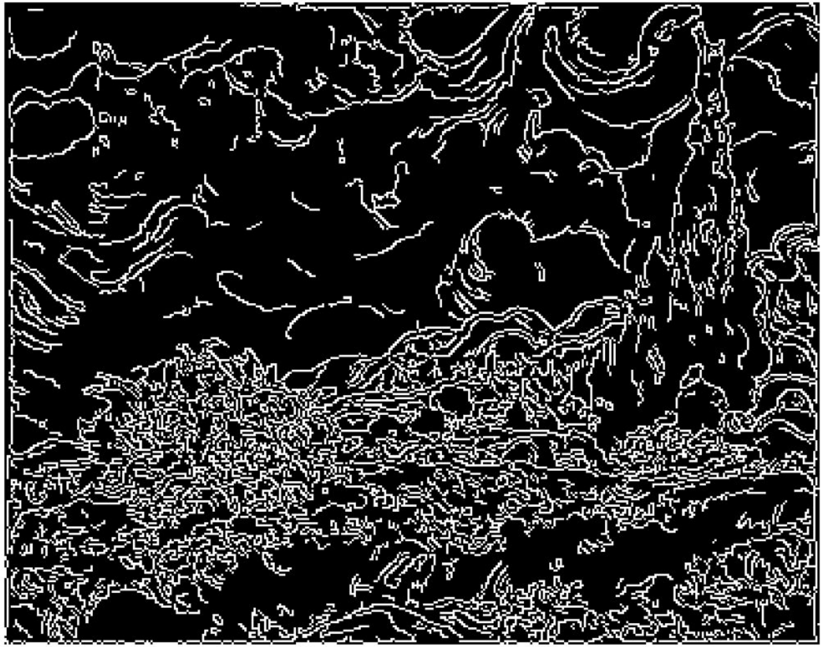
Experiments:

- **Edge-Based Initialization**: Places more points in regions with high color variation (edges)
- **Region-Based Color Averaging**: Starts with average colors in local regions

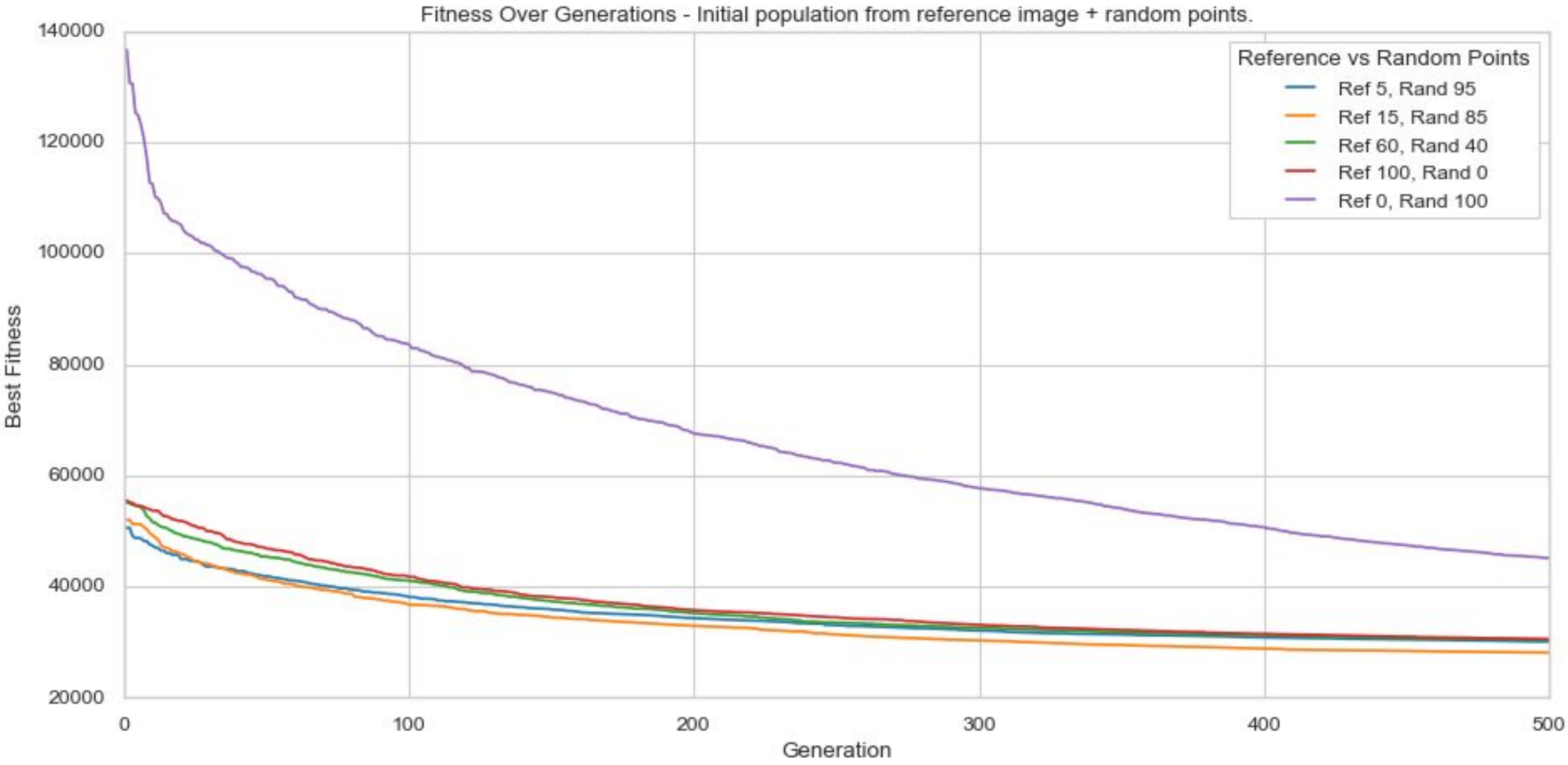
Experiment - 1 Edge Based Initialization

Resized Reference Image

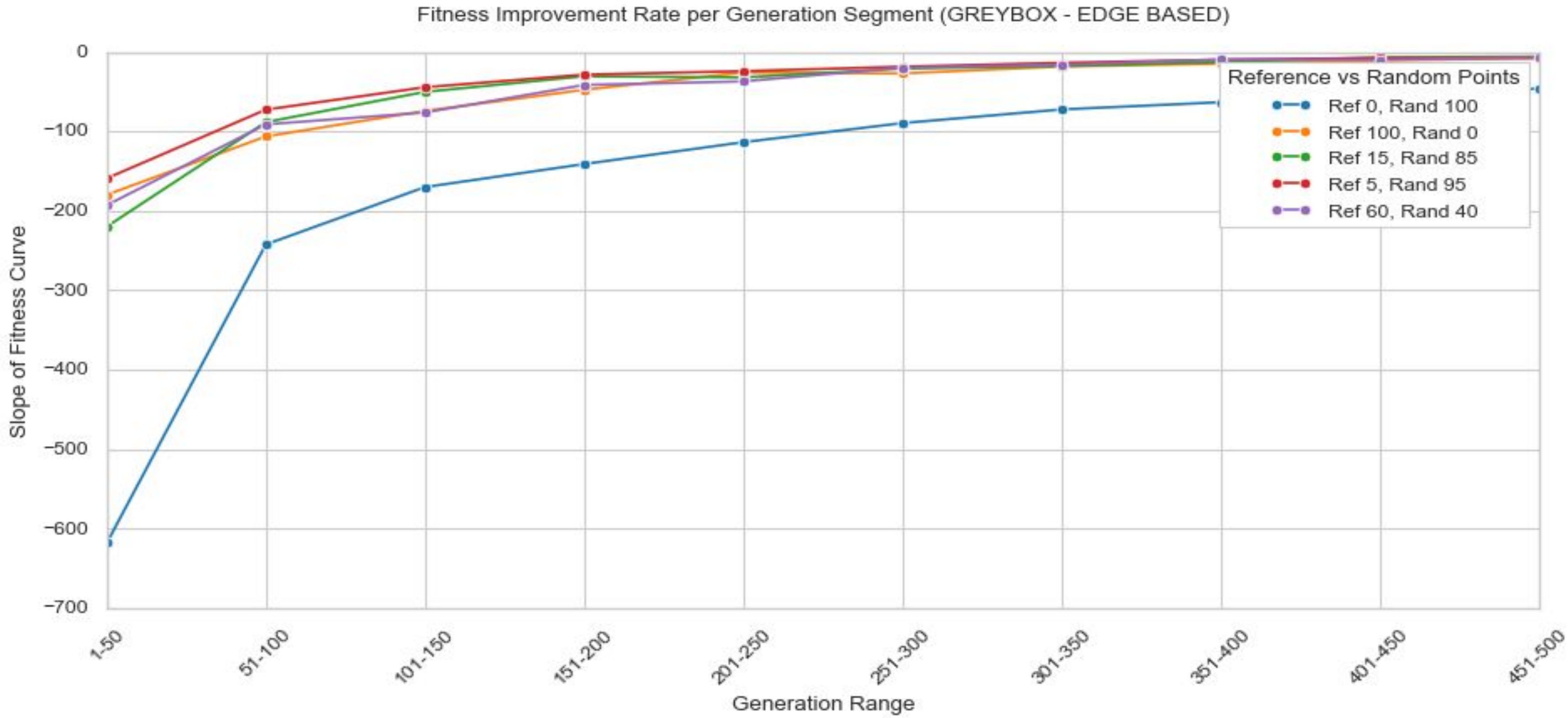
Edges Detected (Canny)



Edge-based initialization: Results

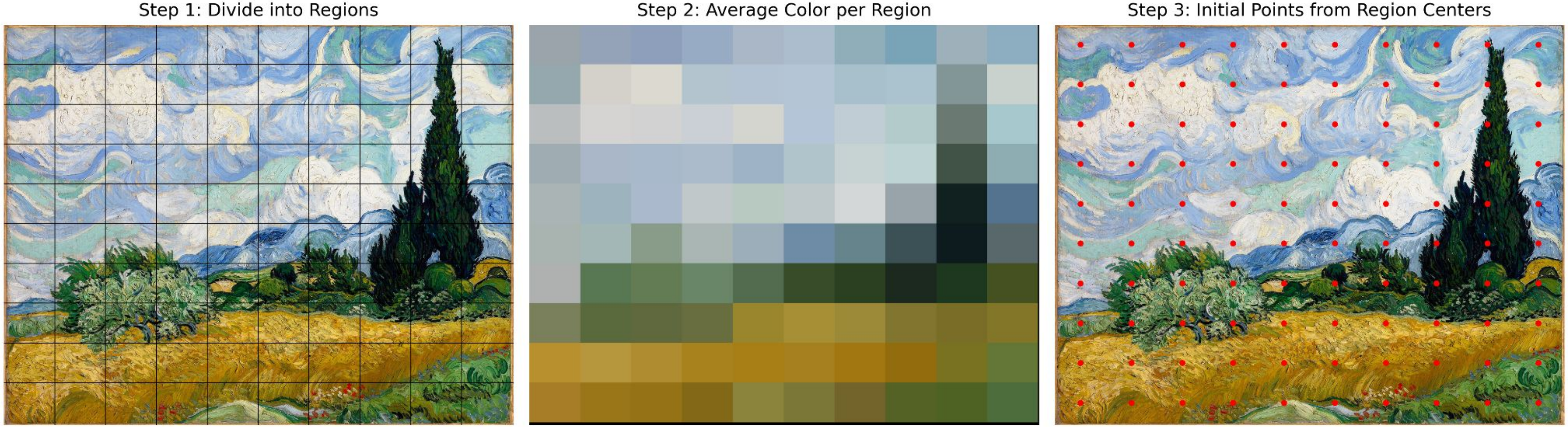


Edge-based initialization: Results

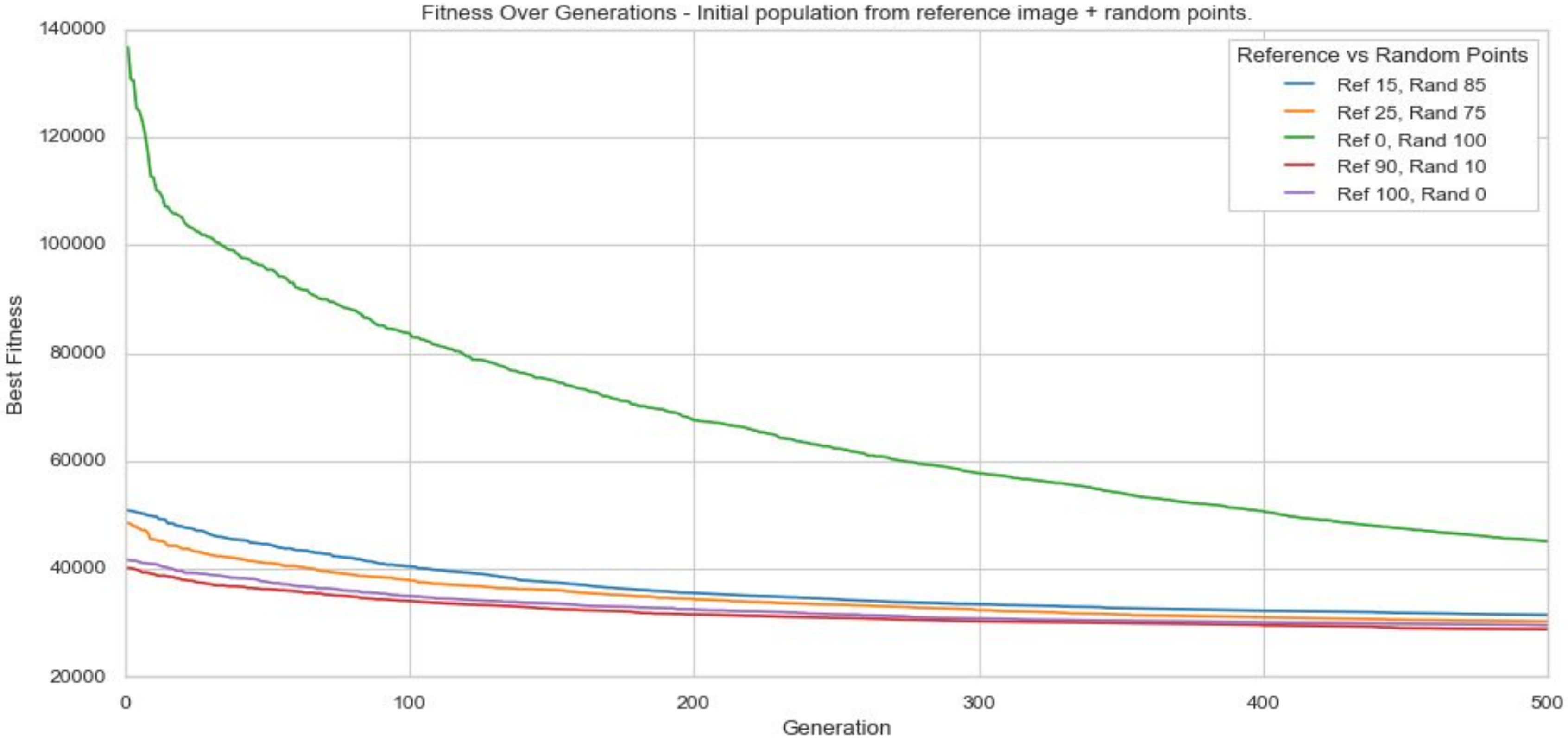


Experiment 2: regions-based

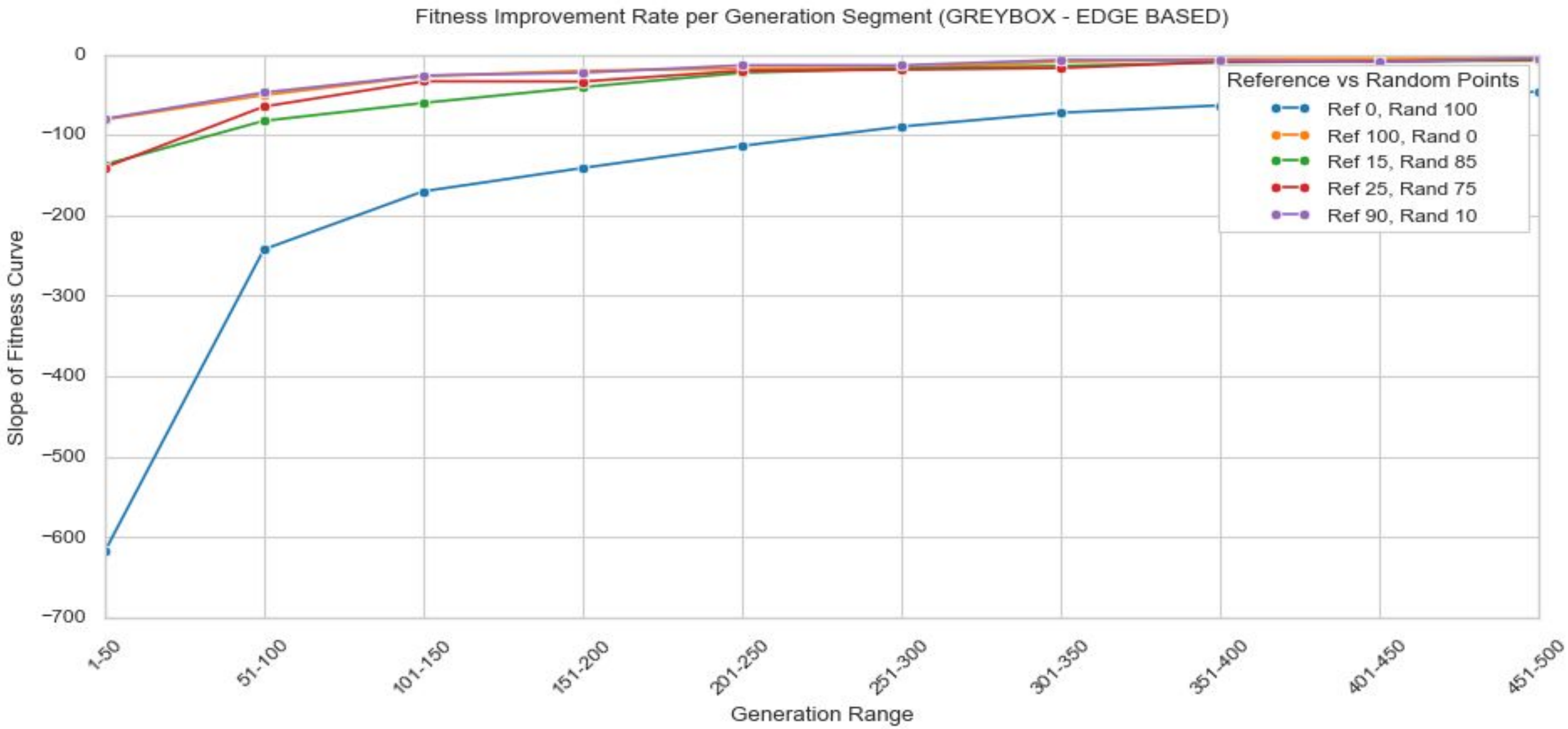
- Captures the image's overall tone



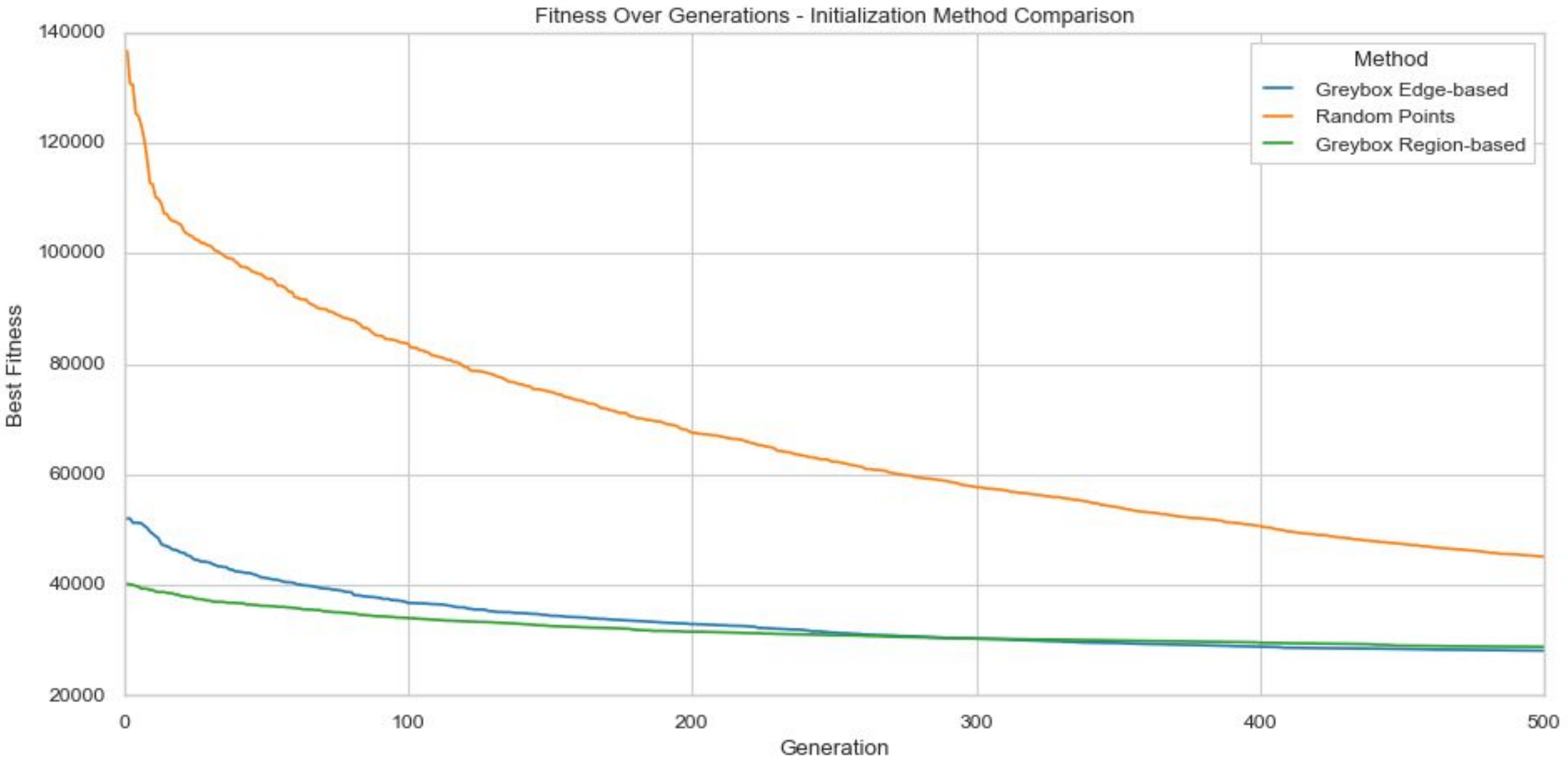
Region-based initialization: Results



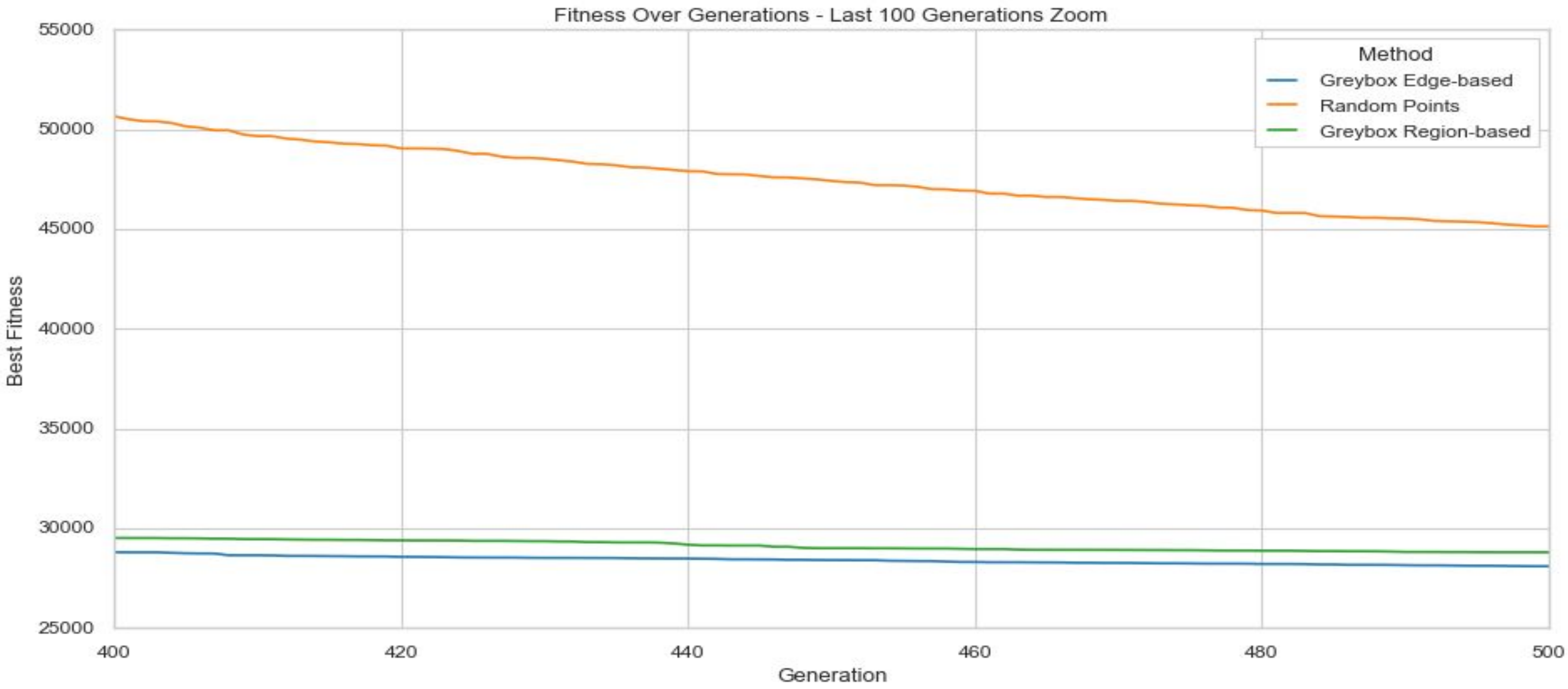
Region-based initialization: Results



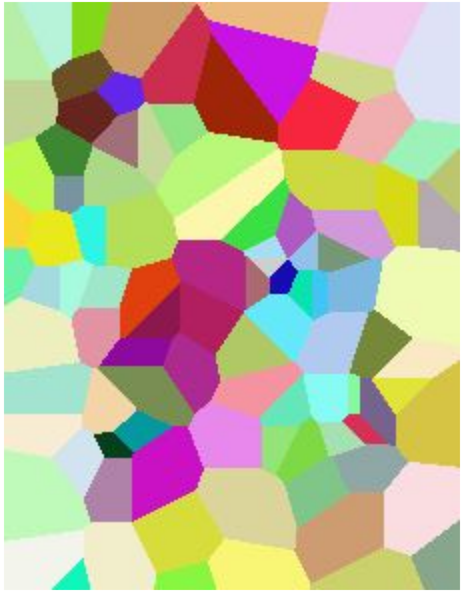
A comparison of the best fitnesses



Last 100 generations

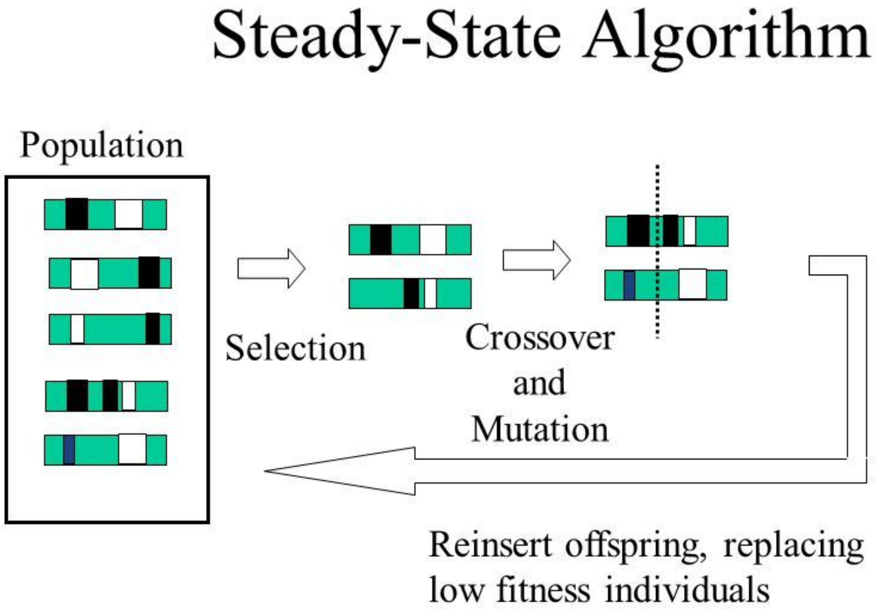


Questions?



Extra slides

Selection Methods to improve avg-fitness : SSGA



Distribution of work

Roxana	Grey-box Initialization: Region-based
Sanchita	Selection-truncation + SSGA
Sreevidya	Grey-box Initialization: Edge-based
Sofie	Selection-tournament + SUS
Tygo	Voronoi-preserving Crossover

Preserving Voronoi-points during Variation: Goal

Goal of variation: We want to combine good schema (building blocks) of well-performing parents to get even better offspring

Schema growth equation: Schema’s with an above average fitness are expected to appear more often in the population when their probability of being disrupted is sufficiently low

Thus: Reducing the probability of disruption will encourage schemata with above average fitness

Preserving Voronoi-points during Variation: Idea

What does a building block in a Voronoi diagram look like?

Probably: a point or set of points that captures the image well

Because: A good schema is unlikely to contain only a part of a point, because the values (x, y, r, g, b) are dependent: it makes little sense for an “xy”-value to be good on its own. Instead: “Color rgb at xy is good” makes more sense

So: Keeping individual Voronoi points intact during crossover will reduce the probability of disruption

Let’s design crossover operators that work on individual **points** rather than individual **integers**

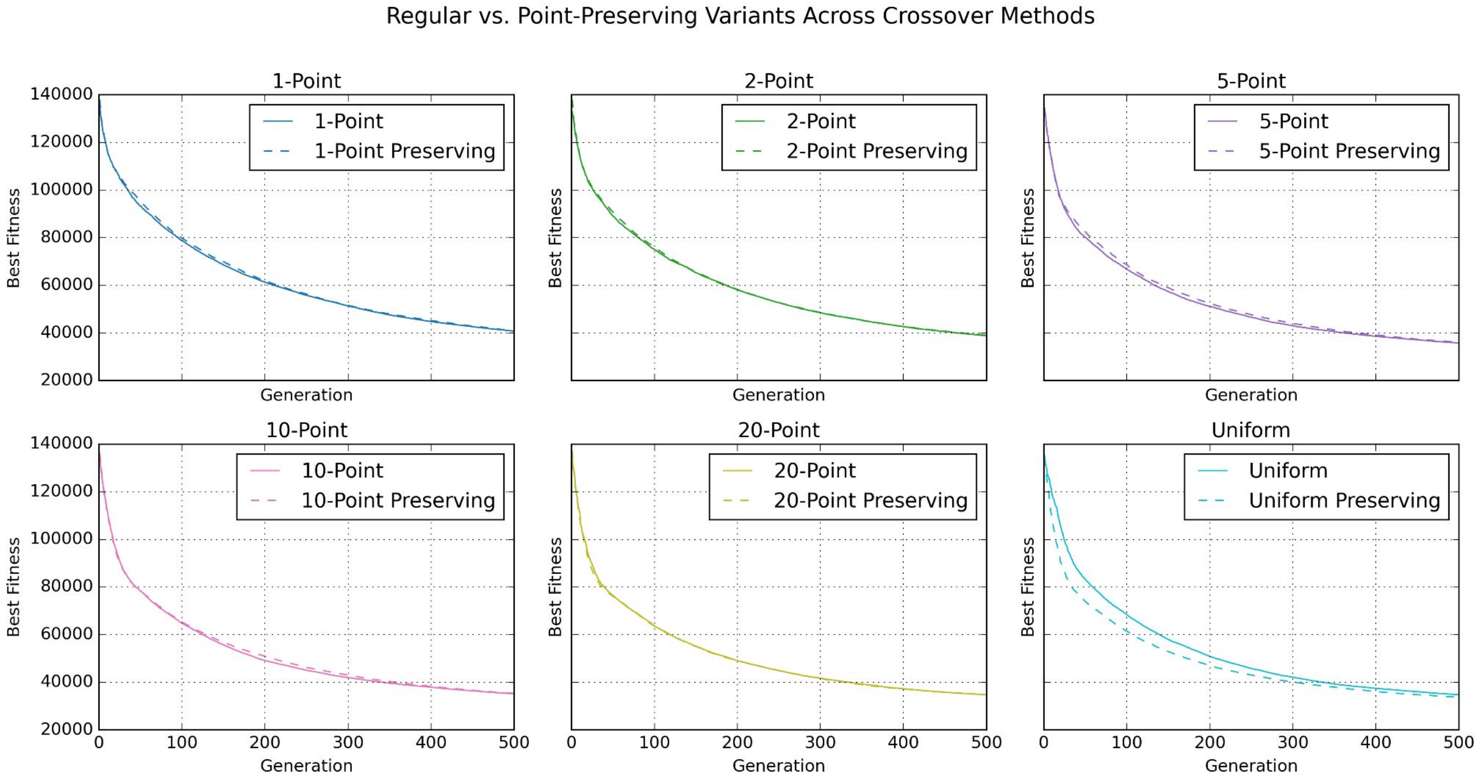
Preserving Voronoi-points during Variation: Experiment

RQ: How does preserving the (x, y, r, g, b)-structure of individual Voronoi points during crossover affect the algorithm's convergence rate?

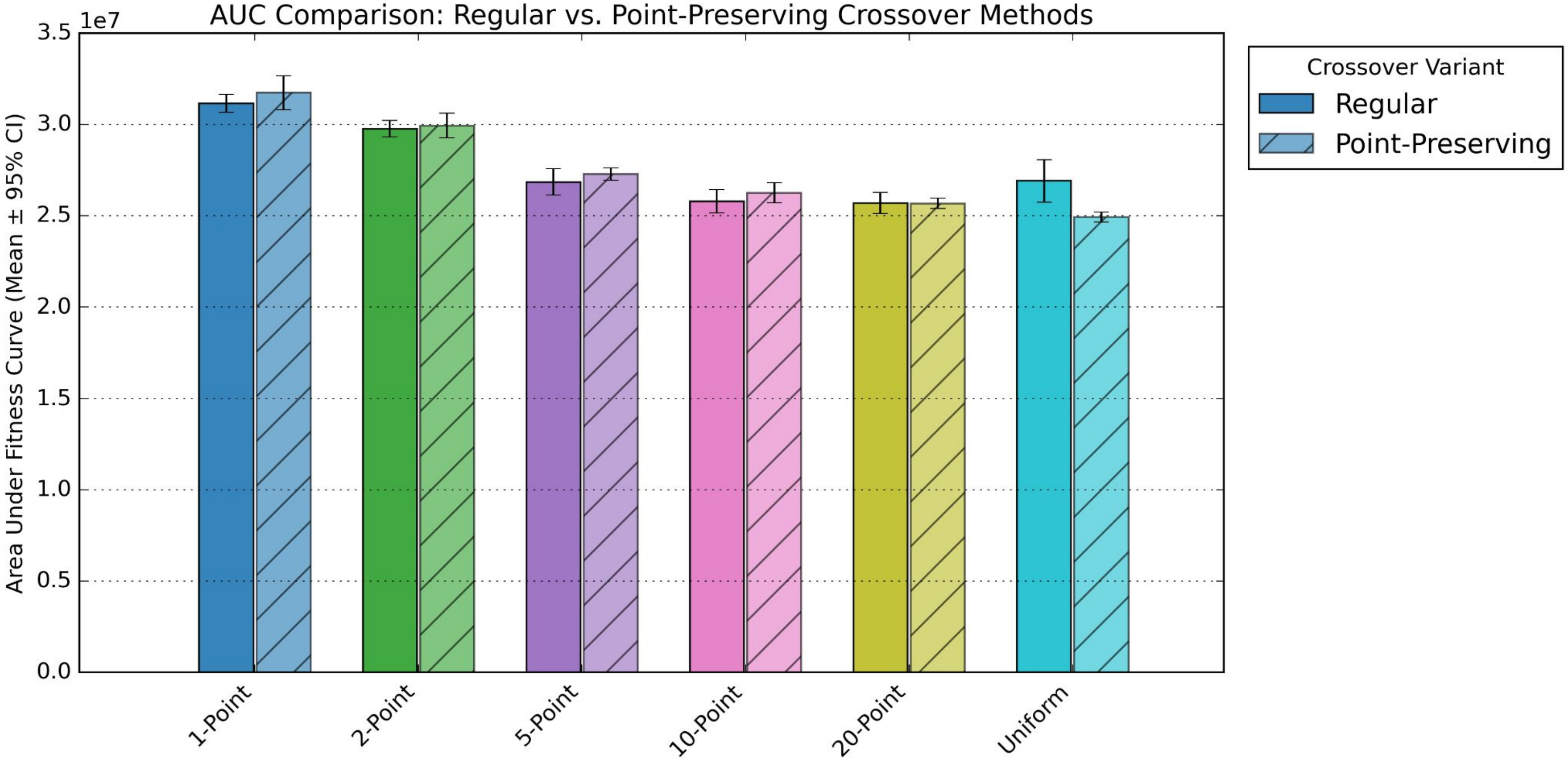
To find out, we take several common crossover-operators and implement a version that works on integers and one version that works on points

Then, we measure the best fitness over generations per operator, and we compare the point-based operators to their integer-based counterparts

Preserving Voronoi-points during Variation: Results



Preserving Voronoi-points during Variation: Results



Preserving Voronoi-points during Variation: Conclusion

Point-preserving crossover does not improve n-point crossover for n<=20

Point-preserving uniform crossover seems to perform the best

Possible reasons:

- n-point crossover already does not disrupt blocks that much (for low n)
- point-preserving crossover might not introduce enough variation

Tournament sizes